Part I: Unix Review and File Setup.

If you are on windows, you will *need* to install git for this part of the lab. This will allow you to use a shell terminal called *"Git Bash"* locally on your computer. For Mac users, the default "Terminal" application works great.

Consider the following file system on a fellow Berkeley student's computer. We are looking at the home directory `"~"`:

```
~
├── Documents
|   ├── econ100b
|   ├── stat134
|   |   ├── syllabus.pdf
|   |   └── hw
|   └── data100
├── Downloads
|   ├── image.png
|   ├── download.zip
|   ├── hw1.pdf
|   └── f.exe
...
```

1. Write the relative file path starting at the "Downloads" directory to the "hw" directory in "stat134".

2. Write the absolute path of "hw1.pdf".

3. Use the `"mv"` command to move the "hw1.pdf" file from Downloads to the "hw" directory in "stat134". You may use relative or absolute paths.

```
$

$
```

Now for some sanity checks. Open your computer's Terminal (or Git Bash) and navigate to the home directory "`~`". This is typically the default directory when you open either application.

For mac users, when you run `pwd` you should see:

```
/Users/[username]
```

For windows running `pwd`:

```
/c/Users/[username]
```

At this point, if you run `ls` you should see `Downloads`, `Documents`, `Desktop`, and more.

4. Create a `stat133` folder within your `~/Documents` directory.

```
$

$

$
```

5. Download the `data-in-trees` github repository locally. This will be packaged as a `zip` file. Unzip it to your `Downloads` folder. Then copy the new unzipped folder into your new `stat133` directory. This will default the name `data-in-trees-master`.

```
$

$

$
```

6. *Challenge*: Display the entire tree of contents for `data-in-trees-master` with a single line.[1]

```
$
```

[1] Hint: This is known as *recursive* listing. You may reference the ls manual

Part II: First Contact with R.

You should have the R App installed on your computer. If not, navigate to The R Project for Statistical Computing website and download it from any CRAN mirror (https://cloud.r-project.org works just fine).

7. Set the working directory of R session to be the `stats` directory for your current time at berkeley's folder. For example, if you've been a berkeley student for 1 year, your working directory should be: `"~/Documents/stat133/data-in-trees-master/stats/one"`.

```
>

>
```

8. Use a `for` loop to fill the vector `"contents"` with the data on each notecard. Your final vector should follow the pattern: c(song, distance, song, distance, ...).

```
> contents <- c()

>

>

>

>

>

>

>
```

9. Fill the matrix `"mat"` with each row corresponding to one student, and each column corresponsing to the two questions answered.

```
> mat <-
```

10. **Using a for loop**, fill the vector `"far_distance"` with logicals which hold `TRUE` if the student traveled **more than** 1,000 miles over the summer, and `FALSE` **otherwise**. How many responses hold TRUE?

```
> far_distance <- c()

>

>

>

>

>

>

>

>

>

>
```

11. **Using recycling**, fill and define a vector `"far_distance"` with logicals which hold `TRUE` if the student traveled more than 1000 miles over the `summer`, and `FALSE` otherwise. How many responses hold `TRUE` (is it the same is 7)?

```
>

>

>

>
```

12. *Challenge:* Of the different methods you used in the previous two problems, which process do you suspect is faster? Why?

13. Is it possible to fill the second column of `"mat"` with numeric values instead of string values? Why or why not?